

Contents

CerviQ Training and Evaluation	1
Build	1
Final V4 training recipe	1
Training environment	2
Configurable training	2
Pairwise evaluation	2
Evaluation methodology	2
Evaluation results	3
1. Learned-policy progression	3
Interpretation	3
2. Deterministic fallback ablation	3
Last-standing by scenario	3
Macro-average supporting metrics	4
3. Final policy against heuristic opponents	4
4. Held-out Large Cave evaluation	4
5. Supplementary direct learned-agent evaluation	5
V4 Reformed vs V3 Diverse	5
Final-model self-play	5
6. Overall experimental conclusions	6
Local report corpus	6
Interpretation limits	7

CerviQ Training and Evaluation

This document records CerviQ's training and evaluation methodology together with results from the final benchmark runs. The command-line tools use the normal game engine and the original two-worm scenario states. They are intentionally separate from the adaptive multi-worm food and roster behavior of the Gloss application.

The tools use process-global randomness and have no seed option. Commands with the same arguments repeat the methodology, but not necessarily the exact episode sequence or numerical result.

The quantitative tables below were derived from 41 aggregate reports retained locally under `outputs/evaluation/` when this document was prepared. The repository ignores `outputs/`, so those raw reports are not distributed by Git. The tables document the recorded runs, but a clean clone cannot independently audit their source values unless the reports are archived separately.

Build

From the repository root:

```
stack build
```

Run experiment commands from the repository root because registered agents and direct model references use repository-relative paths.

Final V4 training recipe

The dedicated command preserves the source-level final V4 configuration:

```
stack run cerviq-train-v4
```

`app/TrainV4.hs` fixes 30,000 episodes, a 500-tick limit, alpha 0.1, gamma 0.9, epsilon 1.0 down to 0.05 with decay 0.99985, and progress every 200 episodes. It writes:

```
data/models/qlearning_v4_reformed_diverse_30k.txt
data/models/qlearning_v4_reformed_diverse_30k_stats.txt
```

The final recipe starts from an empty Q-table. It is kept unchanged as the historical recipe used to produce the final V4 model.

Training environment

The final diverse training environment samples from the following distributions:

Dimension	Distribution
Scenario	Arena 40%, Cave 30%, Corridors 30%
Opponent	Safe Greedy Food 40%, Safe Hunter 30%, Safe Random 20%, Random 10%
Starting side	Original 50%, first two bodies/directions swapped 50%
Food	Authored food cleared, then one ordinary random item

Only each scenario's two default worms are used. Large Cave and the additional interactive multi-worm starts are not part of this training distribution.

Configurable training

`cerviq-train` can train any implemented state version:

```
stack run cerviq-train -- VERSION [OPTIONS]
```

VERSION is v1, v2, v3, or v4. The exact supported options can be shown with:

```
stack run cerviq-train -- --help
```

The configurable trainer is primarily intended for shorter experiments and configurable demonstrations. The dedicated `cerviq-train-v4` command remains the record of the final 30,000-episode V4 recipe.

Pairwise evaluation

A registered agent can be evaluated against another registered agent with:

```
stack run cerviq-evaluate -- \
  --agent "Q-learning V4 Reformed 30k + fallback" \
  --opponent "Safe Greedy Food" \
  --scenario Arena \
  --episodes 1000 \
  --max-ticks 500
```

Useful discovery commands are:

```
stack run cerviq-evaluate -- --list-agents
stack run cerviq-evaluate -- --list-scenarios
```

The evaluator also accepts direct Q-table references using the prefixes `v1:`, `v2:`, `v3:`, `v3f:`, `v4:`, and `v4f:`. The `f` variants use the corresponding fallback policy.

Evaluation methodology

Every benchmark in the local main evaluation set uses the same setup:

1. construct `Scenario.scenarioInitialState`, containing the two default worms;
2. randomly keep or swap the two starting bodies and directions;
3. clear authored food and place exactly one ordinary random food item;
4. assign the selected controllers to Worms 1 and 2;
5. simulate for at most 500 ticks;
6. stop earlier only if all worms are dead.

All main benchmark reports and all direct learned-agent comparisons use **1,000 episodes per matchup and scenario**. Final-model self-play is stored as two independent 500-episode shards per scenario and combined as a 1,000-episode symmetry check. The start side and food location are re-randomized for every episode.

The main performance measure in the tables below is **last-standing rate**. This is not called a win rate: a worm counts as last-standing if it becomes the only living worm at any point, even if it later dies. Final survival rate is therefore reported separately where useful.

When a single mean is shown across Arena, Cave, and Corridors, it is an **unweighted macro-average across the three scenarios**. Large Cave is kept separate because it was not part of the training distribution.

The local reports contain aggregate statistics rather than per-episode traces. The comparisons below are therefore descriptive; no significance claim is made for small differences between stochastic runs.

Evaluation results

1. Learned-policy progression

The primary progression experiment compares successive **raw learned policies without fallback** against the same **Safe Greedy Food** reference opponent. This keeps the opponent and evaluation protocol fixed while the learned representation and model change.

Model	Arena LS	Cave LS	Corridors LS	Mean LS	Mean survival	Mean age	Food / 100 ticks
V1	48.2%	47.4%	28.9%	41.5%	24.5%	229.4	3.39
V2	70.5%	76.1%	50.4%	65.7%	61.7%	392.9	3.49
V3 Diverse 30k	73.0%	79.5%	48.2%	66.9%	49.6%	377.7	3.53
V4 Diverse 30k	76.0%	85.1%	69.0%	76.7%	40.4%	366.5	4.03
V4 Reformed 30k	78.0%	87.1%	71.6%	78.9%	51.3%	387.1	4.33

Interpretation

The largest early improvement is from V1 to V2: mean last-standing rises from 41.5% to 65.7%, with substantial gains on all three maps. V2 also survives much longer on average.

V3 Diverse is a more mixed step. It improves last-standing on Arena and Cave but falls from 50.4% to 48.2% on Corridors, producing only a small macro-average gain over V2. Its final survival rate is also lower than V2's. This is useful evidence that later state representations did not improve every metric monotonically.

V4 Diverse produces a clearer competitive improvement, especially on Corridors, where last-standing rises from 48.2% for V3 to 69.0%. Its mean last-standing rate reaches 76.7%. At the same time its final survival rate is lower than V3's. The combination demonstrates why last-standing and final survival should be treated as different metrics: a policy can more often become the sole survivor without necessarily remaining alive until the tick limit.

V4 Reformed gives a small but consistent last-standing improvement over V4 Diverse on all three training maps (78.0%, 87.1%, and 71.6%). It also recovers some of the lost survival and age while achieving the highest raw-policy food rate. The 2.2 percentage-point macro last-standing increase over V4 Diverse is small enough that it should be treated as descriptive rather than as a strong statistical claim by itself.

2. Deterministic fallback ablation

V3 and V4 can combine their learned Q-table with a deterministic safety/topology fallback. To isolate its practical effect, the same learned table is evaluated with and without fallback against **Safe Greedy Food**.

Last-standing by scenario

Model	Arena raw $\rightarrow$ fallback	Cave raw $\rightarrow$ fallback	Corridors raw $\rightarrow$ fallback	Mean change
V3 Diverse 30k	73.0% $\rightarrow$ 81.1%	79.5% $\rightarrow$ 80.0%	48.2% $\rightarrow$ 49.0%	+3.1 pp
V4 Diverse 30k	76.0% $\rightarrow$ 91.7%	85.1% $\rightarrow$ 93.1%	69.0% $\rightarrow$ 81.5%	+12.1 pp
V4 Reformed 30k	78.0% $\rightarrow$ 88.6%	87.1% $\rightarrow$ 92.2%	71.6% $\rightarrow$ 82.6%	+8.9 pp

Macro-average supporting metrics

Model	Raw LS	+ fallback LS	Raw survival	+ fallback survival	Raw age	+ fallback age	Raw food/100
V3 Diverse 30k	66.9%	70.0%	49.6%	58.0%	377.7	401.8	3.53
V4 Diverse 30k	76.7%	88.8%	40.4%	89.3%	366.5	462.3	4.03
V4 Reformed 30k	78.9%	87.8%	51.3%	88.1%	387.1	456.3	4.33

The fallback has only a modest mean last-standing effect on V3, but a much larger effect on both V4 tables. For V4 Diverse it adds 12.1 percentage points of mean last-standing and almost 49 percentage points of final survival. For V4 Reformed it adds 8.9 points of mean last-standing and 36.8 points of survival.

The safety gain is not free in every metric. Mean food per 100 ticks decreases for both V4 tables when fallback is enabled. The final runtime policy should therefore be described as a combination of learned Q-values and deterministic safety/topology handling, rather than attributing all of its benchmark strength to the learned table alone.

One non-monotonic result should be kept visible: under this particular **Safe Greedy Food** benchmark, **V4 Diverse + fallback has the highest mean last-standing rate (88.8%)**, slightly above V4 Reformed + fallback (87.8%). The raw V4 Reformed policy is stronger than raw V4 Diverse in these runs, but adding the fallback does not preserve that ordering exactly.

3. Final policy against heuristic opponents

The selected final interactive policy, **Q-learning V4 Reformed 30k + fallback**, was evaluated against both heuristic opponents that are most useful as strong references.

Opponent	Arena LS	Cave LS	Corridors LS	Mean LS	Mean survival	Mean age	Food / 100 ticks
Safe Greedy Food	88.6%	92.2%	82.6%	87.8%	88.1%	456.3	3.67
Safe Hunter	60.4%	86.0%	61.6%	69.3%	69.3%	370.6	3.83

Safe Hunter is clearly the harder matchup in Arena and Corridors. Against it, mean last-standing falls by about 18.5 percentage points relative to Safe Greedy, and final survival falls by about 18.8 points. Cave remains the strongest map for the final policy, with an 86.0% last-standing rate even against Safe Hunter.

These heuristic comparisons are the primary reference benchmark because every learned model can be compared against the same fixed opponent policy. Direct Q-learning-versus-Q-learning comparisons are treated as supplementary evidence below.

4. Held-out Large Cave evaluation

Large Cave is excluded from the mixed training scenario distribution, so it is reported separately as a held-out environment. The final policy was evaluated against both heuristic references.

Opponent	Agent LS	Agent survival	Opponent LS	Opponent survival	Agent avg age	Agent food/100	Avg
Safe Greedy Food	31.8%	91.5%	1.7%	61.3%	480.5	1.83	
Safe Hunter	72.2%	74.4%	1.1%	3.3%	408.6	2.30	

The 31.8% last-standing result against Safe Greedy should **not** be read as only 31.8% survival. The final policy survives to the end in 91.5% of Large Cave episodes, while Safe Greedy also survives in 61.3%. The average episode lasts 482.5 of the allowed 500 ticks. On this much larger map, many episodes therefore finish at the tick limit without either worm ever becoming the sole survivor. This makes last-standing less directly comparable with the smaller training maps.

Against Safe Hunter, the final policy reaches 72.2% last-standing and 74.4% final survival while the opponent survives only 3.3% of episodes. Together the two Large Cave runs show that the final policy can operate successfully on the held-out geometry, but they do not establish improvement over older learned models because those models were not evaluated on Large Cave in this benchmark set.

5. Supplementary direct learned-agent evaluation

The primary progression deliberately uses fixed heuristic opponents. Direct learned-agent tests complement that benchmark by placing two learned policies in the same episodes. These comparisons remain supplementary because they do not provide the common fixed reference used in Sections 1--4.

V4 Reformed vs V3 Diverse

Q-learning V4 Reformed 30k + fallback was evaluated directly against Q-learning V3 Diverse 30k + fallback on Arena, Cave, and Corridors. Each map was evaluated twice for 1,000 episodes: once with V4 Reformed assigned to Worm 1 and once with the controller assignments reversed. Physical starting sides were still randomized independently every episode.

Scenario	V4 LS as W1	V4 LS as W2	V3 LS as W1	V3 LS as W2	V4 seat-avg LS	V3 seat-avg LS	Differ
Arena	67.4%	68.2%	29.5%	30.9%	67.8%	30.2%	+37.6%
Cave	63.1%	61.6%	24.3%	25.7%	62.4%	25.0%	+37.4%
Corridors	37.3%	35.6%	24.7%	27.3%	36.5%	26.0%	+10.5%
Macro average	55.9%	55.1%	26.2%	28.0%	55.5%	27.1%	+28.5%

Supporting metrics averaged over both controller assignments show the same overall pattern:

Metric	V4 Reformed + fallback	V3 Diverse + fallback
Macro last-standing	55.5%	27.1%
Macro survival	71.3%	34.6%
Mean age	425.8	351.6
Food / 100 ticks	3.34	3.67

V4 Reformed therefore has a large direct competitive advantage over V3 Diverse in Arena and Cave and a smaller but still positive advantage in Corridors. The last-standing gap remains large in both controller assignments, so the conclusion is not caused by choosing one fixed Worm-1/Worm-2 assignment. V4's own last-standing rate changes by only 0.8 percentage points in Arena, 1.5 points in Cave, and 1.7 points in Corridors when the assignments are reversed.

The food-rate result is deliberately kept visible: V3 has a slightly higher macro food-per-100-ticks value, driven mainly by Arena, despite much lower survival and last-standing. Food collection efficiency is therefore not a substitute for competitive survival in this environment.

Final-model self-play

The final policy was also evaluated against itself as a symmetry and sanity check. Each scenario consists of two independent 500-episode shards. Because the shards have equal size, the combined values below are their simple averages and represent 1,000 episodes per scenario.

Scenario	Worm 1 LS	Worm 2 LS	LS difference	Worm 1 survival	Worm 2 survival	Avg episode length
Arena	43.2%	35.6%	+7.6 pp	64.1%	56.8%	499.6
Cave	39.4%	36.9%	+2.5 pp	63.1%	60.6%	500.0
Corridors	40.4%	36.5%	+3.9 pp	63.5%	59.6%	500.0

The two identical controllers behave broadly similarly, but Worm 1 has a consistent descriptive advantage in all three self-play scenarios. The largest gap is 7.6 percentage points of last-standing in Arena. Because physical starting sides are randomized, this cannot simply be attributed to always receiving one particular map side. It may reflect a remaining worm-ID/update-order asymmetry or stochastic sampling variation. The aggregate reports are not sufficient to distinguish those explanations statistically, so this is recorded as a symmetry diagnostic rather than a performance claim.

Self-play also illustrates the effect of the 500-tick limit. Average episode length is essentially 500 ticks on every map, and the two last-standing rates do not sum to 100%. Many episodes therefore end while both identical policies are still alive.

Importantly, the modest self-play asymmetry does not explain the direct V4/V3 result: that experiment explicitly reverses the controller assignments, and V4 Reformed remains substantially ahead of V3 Diverse in both assignments.

6. Overall experimental conclusions

The benchmark results support the following descriptive conclusions:

- V2 provides the largest early improvement over V1, while V3 is a mixed step rather than a uniform improvement on every map and metric.
- V4 produces the clearest later competitive gain, particularly on Corridors. V4 Reformed improves the raw V4 policy modestly and consistently in the three main scenarios.
- The deterministic fallback is a substantial part of practical V4 performance. It strongly increases survival and last-standing, while sometimes reducing food collected per unit time. The final controller must therefore be described as a hybrid learned-plus-deterministic policy.
- The selected V4 Reformed + fallback policy performs strongly against Safe Greedy Food and remains competitive against the stronger Safe Hunter baseline.
- On held-out Large Cave, high survival together with frequent 500-tick timeouts shows why last-standing alone can understate successful long-lived behaviour on large maps.
- Direct learned-agent evaluation confirms a substantial V4 Reformed advantage over V3 Diverse even when controller assignments are reversed.
- Final-model self-play is broadly symmetric but exposes a modest consistent Worm-1 advantage that should be documented as a remaining diagnostic rather than hidden.

These conclusions are descriptive rather than claims of statistical significance because the evaluator stores aggregate reports and uses unseeded randomness.

Local report corpus

The local evaluation corpus used to derive these tables contains **41 report files**:

- 15 raw-policy progression reports: V1, V2, V3 Diverse, V4 Diverse, and V4 Reformed against Safe Greedy Food on Arena, Cave, and Corridors;
- 9 fallback reports: V3 Diverse, V4 Diverse, and V4 Reformed with fallback against Safe Greedy Food on the same three maps;
- 3 final-policy reports against Safe Hunter on Arena, Cave, and Corridors;
- 2 held-out Large Cave reports, against Safe Greedy Food and Safe Hunter;
- 6 direct learned-agent reports comparing V4 Reformed + fallback with V3 Diverse + fallback in both controller assignments on all three training maps;
- 6 final-model self-play shard reports, two 500-episode shards for each of Arena, Cave, and Corridors.

The first 35 files are 1,000-episode reports. The six self-play files are 500-episode shards that combine into three additional 1,000-episode logical evaluations. Thus the corpus represents **38 logical 1,000-episode evaluations** when each self-play shard pair is treated as one experiment.

Interpretation limits

- Training and the main benchmark are one-versus-one even though interactive runtime code supports more opponents.
- The final V4 model was trained only in two-worm environments. Multi-opponent runtime support must not be described as multi-opponent Q-learning training.
- Large Cave is a held-out map and is not included in the three-map training-set macro averages.
- Randomness is unseeded, so repeating a command produces a new stochastic sample rather than the identical episode sequence.
- V3/V4 + **fallback** policies include deterministic safety filters/rankings in addition to learned Q-values; they are not pure greedy Q-table policies.
- Last-standing and final survival measure different outcomes and should not be used interchangeably.
- The local benchmark reports contain aggregate statistics, so the current analysis is descriptive rather than a per-episode statistical significance study.
- Evaluation is simulation-based benchmarking, not automated unit testing. Unit tests are run separately through **stack test**.